macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit

Implementation Solution Guide

Version 1.0.0 · 2026-07-06

Contents

Executive Summary — macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit 4

What you get . . . 4

What this delivers . . . 4

The standard deployment model . . . 4

Outcome . . . 5

1. Solution Overview — macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit 5

What this solution is . . . 5

The macOS + Intune signal path (the layered model) . . . 5

The five failure classes (symptoms) . . . 6

The signals you read (and where) . . . 6

The diagnose → resolve method . . . 7

The standard deployment model . . . 7

2. Requirements Guide — macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit 7

2.1 Intune console access (Microsoft Intune admin center) . . . 7

2.2 Microsoft Graph read access (for the scripts) . . . 7

2.3 Administrative roles for macOS management . . . 8

2.4 Apple enrollment vantage point (ABM / ASM) . . . 8

2.5 On-device diagnostic access (the failing Mac) . . . 8

2.6 Device logs and diagnostics . . . 9

2.7 Current enrollment program token and profiles . . . 9

2.8 Supported macOS baseline . . . 9

2.9 A reproducible failure to pilot on . . . 9

Checklist . . . 9

3. Diagnostic Playbook — macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit 10

Step 0 — Classify the symptom and gather context . . . 10

Branch A — ADE not assigned / enrollment stuck . . . 10

Branch B — SSO won't sign in (Platform SSO / Extensible SSO) . . . 11

Branch C — SecureToken / FileVault error . . . 12

Branch D — Compliant but denied . . . 13

Branch E — User locked out after enrollment . . . 13

Step 9 — Record the resolution . . . 14

Step 10 — Validate and expand . . . 14

5. Testing Guide — macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit
Test 1 — Device context is visible (Discovery) . . . 15
Test 2 — Enrollment / ADE state (Branch A) . . . 15
Test 3 — SSO extension state (Branch B) . . . 15
Test 4 — SecureToken / FileVault state (Branch C) . . . 15
Test 5 — Compliance / Conditional Access (Branch D) . . . 15
Test 6 — Fleet report and escrow coverage . . . 15
Test 7 — Safe remediation preview (default -WhatIf) . . . 16
Test 8 — Scope targeting preview (default -WhatIf) . . . 16
Test 9 — Re-check after fix (any branch) . . . 16
Pass/fail gate . . . 16
6. Troubleshooting Guide — macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit 16
A. Microsoft Graph sign-in / scope errors . . . 17
B. Device not found by the scripts . . . 17
C. On-Mac commands are denied or "command not found" . . . 17
D. Profile shows Error/Conflict in Intune but you can't tell why . . . 17
E. Recovery keys / escrow reads empty . . . 18
F. "Compliant but denied" doesn't resolve . . . 18
G. Remediation / targeting "made no change" . . . 18
H. Diagnostics quick reference . . . 18
I. When to escalate . . . 19
7. Best Practices — macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit 19
8. Deliverables Summary — macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit 20
Documentation . . . 20
PowerShell scripts (Microsoft Graph SDK) . . . 21
What the buyer receives . . . 21
Outcome . . . 21
Appendix A — PowerShell Scripts 22

Executive Summary — macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit

White-label · Infrastructure Solutions

When a Mac won't finish enrolling, won't sign in through Platform/Extensible SSO, throws a SecureToken/FileVault error, is marked compliant yet still blocked, or locks a user out after enrollment, the cause hides across three systems at once — the **Apple** enrollment profile, the **Intune** policy, and the **Entra** identity. Technicians burn hours guessing which layer is at fault, and too many stuck Macs get wiped when they needed a one-line fix.

This solution is a structured, symptom-driven **diagnostic playbook** plus tooling for the macOS + Microsoft Intune stack. It models the signal path as layers — **ADE/enrollment** → **configuration profiles** → **SSO** → **SecureToken/FileVault** → **compliance/Conditional Access** — and gives you a root-cause tree that classifies the symptom, confirms each layer **top-down**, and reads the one authoritative signal per layer, so you isolate the failing layer in minutes. It is written for a competent Intune administrator or Mac technician who needs a proven method instead of trial and error.

What you get

• A plain-language **Solution Overview** of the macOS-Intune signal path, the layered model behind the root-cause tree, the five failure classes, and the diagnose-then-resolve method.

• A **Requirements Guide** with verify / implement-if-missing for every prerequisite — Intune and Microsoft Graph read access, the Apple Business/School Manager vantage point, on-Mac diagnostic access, and device logs.

• A **Diagnostic Playbook** — the symptom tree with a branch for each class (ADE not assigned / enrollment stuck, SSO won't sign in, SecureToken/FileVault error, compliant-but-denied, user locked out) and the exact checks and fixes per branch.

• Five **PowerShell scripts** (Microsoft Graph SDK) for device discovery, per-layer validation, a fleet health and escrow-coverage report, safe Sync-based remediation, and a scope/target-group builder — read-only where it matters, with the two write-capable scripts defaulting to `-WhatIf`.

• **Testing** and **Troubleshooting** guides with explicit pass/fail gates, plus **Best Practices** for evidence-based escalation.

What this delivers

A diagnostic capability your team keeps: every macOS ticket is triaged the same way — classified, confirmed top-down, and resolved at the layer that actually failed — so Macs stop getting wiped on a hunch and knowledge stops walking out the door. Pilot the playbook on real failures, prove it, then make it the standard.

The standard deployment model

Assess → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

The failing layer of a stuck Mac isolated in minutes — enrollment, SSO, SecureToken/FileVault, or compliance — with a repeatable playbook and scripts you keep and reuse, validated on real failures before you scale it across the support team.

1. Solution Overview — macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit

What this solution is

A structured, symptom-driven method — a diagnostic playbook plus tooling — for troubleshooting the **macOS + Microsoft Intune** stack when a Mac won't finish enrolling, won't sign in through **Platform/Extensible SSO**, throws a **SecureToken/FileVault** error, is compliant but still denied by **Conditional Access**, or locks a user out after enrollment. Instead of wiping the Mac on a hunch, you read the signals the stack already produces — the **Automated Device Enrollment (ADE)** assignment, the delivered **configuration profiles**, the SSO extension state, SecureToken/FileVault status, and the compliance/Conditional-Access result — and follow a root-cause tree that isolates the **one failing layer** in minutes. The result is a repeatable diagnose-then-resolve loop any competent Intune administrator or Mac technician can run.

The macOS + Intune signal path (the layered model)

A working, trusted Mac is the product of a chain of layers, each of which must succeed before the next can. When something fails, it is almost always **one** layer — and the layers fail in a predictable order, so you diagnose top-down:

1. **ADE / enrollment.** The Mac's serial is assigned to your MDM in **Apple Business Manager**

(ABM) or **Apple School Manager** (ASM), which is synced to Intune's **enrollment program token**. During **Setup Assistant** the Mac contacts Apple, is handed to Intune, and enrolls with an **enrollment profile**. If the serial isn't assigned, or the token is expired, or the profile isn't set as default — enrollment never starts.

1. **Configuration profiles.** Once enrolled, Intune delivers the **device configuration profiles**

the later layers depend on: the **SSO app extension** (Platform SSO or the Microsoft Enterprise SSO plug-in / Extensible SSO), the **compliance-supporting** settings, and any **FileVault** policy. Profiles are targeted by group; if the Mac isn't in the right group, or a profile shows **Error/Conflict**, the dependent layer above it can't work.

1. **SSO (Platform SSO / Extensible SSO).** With the extension profile delivered and the

Company Portal (or Microsoft **Authenticator**/**Intune** app for the plug-in) installed and registered, the user signs in with their Entra ID identity. **Platform SSO** additionally binds the local account to Entra. If the extension isn't registered, the app isn't installed, or the method (password / secure-enclave key / smart card) is misconfigured, sign-in fails here.

1. **SecureToken / FileVault.** On Apple Silicon and modern macOS, a user needs a **SecureToken** (or

the volume-owner **Bootstrap Token**, escrowed to Intune) to be enabled for FileVault and to unlock the disk at boot. Intune can escrow the **FileVault personal recovery key**. If the first account never gained SecureToken, or the Bootstrap Token wasn't escrowed, FileVault enablement or a user's disk-unlock fails.

1. **Compliance / Conditional Access.** Intune evaluates the Mac against the **compliance policy**;

the result flows to Entra ID, where **Conditional Access** decides whether the identity on that device may reach a resource. A Mac can be perfectly enrolled and still be **denied** — because it is non-compliant, or because a CA policy (device state, app, grant control) blocks it.

> ***Read the model top-down.*** *A failure at a lower layer (SSO, FileVault, CA) is very often caused by a missing or errored artifact* ***above*** *it (an unassigned serial, an undelivered profile). The playbook always confirms the higher layer before blaming the lower one.*

The five failure classes (symptoms)

Almost every "stuck Mac" ticket is one of these. Each has a diagnostic tell and a branch in the Diagnostic Playbook (§3):

• **ADE not assigned / enrollment stuck.** Setup Assistant shows no **Remote Management** screen, or

enrollment hangs / errors. Tell: the serial isn't assigned to the MDM in ABM/ASM, the enrollment program token is expired, or no default enrollment profile is set.

• **SSO won't sign in.** The Platform/Extensible SSO prompt never appears, loops, or rejects the

Entra credential. Tell: the SSO extension profile isn't delivered/registered, the required app (Company Portal / Authenticator) isn't installed, or the extension configuration is wrong.

• **SecureToken / FileVault error.** FileVault won't enable, a user can't unlock the disk at boot, or

the recovery key isn't escrowed. Tell: the account lacks SecureToken, the Bootstrap Token wasn't escrowed to Intune, or the FileVault profile is in Error.

• **Compliant but denied.** The device shows **Compliant** in Intune yet the user is still blocked

from a resource. Tell: a Conditional-Access policy (device state, sign-in risk, app control) is denying — or the compliance signal hasn't reached Entra yet.

• **User locked out after enrollment.** Enrollment succeeded but the user can't log in / is prompted

for an unknown password, or the local account is out of sync with Entra. Tell: Platform SSO account binding, a FileVault-enabled account mismatch, or a mobile-account/password issue.

The signals you read (and where)

Layer	Primary signal	Where you read it
ADE / enrollment	Serial assignment, token expiry, default profile	ABM/ASM; Intune → **Devices** → **Enrollment** → **Enrollment program tokens**
Profiles	Profile assignment + per-device state (`Succeeded`/`Error`/`Conflict`)	Intune → **Devices** → **Configuration**; device → **Device configuration**

Layer	Primary signal	Where you read it
SSO	Extension delivered/registered; app installed	Device profile state; on-Mac `Profiles` pane / `app-sso` diagnostics
SecureToken / FileVault	SecureToken holder, Bootstrap Token escrow, recovery key escrow	On-Mac `sysadminctl`/`fdesetup`; Intune device → **Recovery keys**
Compliance / CA	Compliance state; CA policy result; sign-in log	Intune device → **Compliance**; Entra → **Sign-in logs** / **Conditional Access**

The diagnose → resolve method

> *1. **Classify the symptom** into one of the five classes. 2. **Confirm the layer above** the symptom is healthy (top-down) — assignment, then profile, then app/extension. 3. **Read the layer's signal** (the table above) to find the failing artifact. 4. **Apply the branch fix** from the Playbook (§3). 5. **Re-check the signal** and confirm the symptom is gone. 6. **Record** the symptom, the failing layer, the cause, and the fix so the next ticket is faster.*

The standard deployment model

> ***Assess** the failing Mac and its symptom class → **Configure** the read access and vantage points (Graph read, device logs, ABM/ASM) → **Pilot** the playbook on real failures → **Validate** the diagnoses and fixes → **Expand** it to the whole support team. Repeatable pattern: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.***

2. Requirements Guide — macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit

Verify **every** requirement before you rely on the playbook. Each entry: **Description · Why · Verify · Implement-if-missing.** Category-wide prerequisites are in `global-requirements-infrastructure.md` — note that the global list is Windows-centric; the macOS-specific vantage points below are what make the difference on a Mac ticket.

2.1 Intune console access (Microsoft Intune admin center)

Description: Sign-in to `https://intune.microsoft.com` on a current tenant, with the workloads for **Devices**, **Enrollment**, **Configuration**, and **Compliance** visible. **Why:** Enrollment tokens, configuration-profile state, compliance results, and recovery-key escrow are all read from the Intune admin center. **Verify:** `https://intune.microsoft.com` loads; **Devices** → **macOS** lists Macs, and **Devices** → **Enrollment** is reachable. **Implement:** Assign a licensed Intune plan (Global Requirements G2) and an admin role (2.3).

2.2 Microsoft Graph read access (for the scripts)

Description: The **Microsoft.Graph.Authentication** PowerShell module and the ability to consent to read-only Graph scopes. **Why:** The Discovery, Validation, and Report scripts query Graph read-only for managed macOS devices, profile states, compliance states, and recovery-key metadata. Remediation and Set-ScopeTargeting additionally need write/privileged scopes. **Verify:** `Get-Module -ListAvailable Microsoft.Graph.Authentication` returns the module; `Connect-MgGraph -Scopes 'DeviceManagementManagedDevices.Read.All'` completes and `Get-MgContext` shows your account. **Implement:** `Install-Module Microsoft.Graph.Authentication -Scope CurrentUser`. Read scopes used: `DeviceManagementManagedDevices.Read.All`, `DeviceManagementConfiguration.Read.All`, `DeviceManagementServiceConfig.Read.All`, `Directory.Read.All`. Write scopes (Remediation/Targeting only): `DeviceManagementManagedDevices.PrivilegedOperations.All`, `Group.ReadWrite.All`, `GroupMember.ReadWrite.All`.

2.3 Administrative roles for macOS management

Description: Intune Administrator (or a scoped custom role granting read on devices/configuration/compliance and the specific actions you use), plus a reader for Entra sign-in logs. **Why:** Reading enrollment/profile/compliance state and issuing safe device actions (Sync) requires elevated Intune rights; diagnosing "compliant but denied" requires reading Entra **Sign-in logs**. **Verify:** Entra → **Roles and administrators** shows your account with **Intune Administrator**; you can open Entra → **Sign-in logs**. **Implement:** Assign **Intune Administrator**; add a **Security Reader** (or Reports Reader) for sign-in logs. Reserve Global Administrator for tenant-wide changes only.

2.4 Apple enrollment vantage point (ABM / ASM)

Description: Access to **Apple Business Manager** (or Apple School Manager) for the organization, or a named contact who has it. **Why:** "ADE not assigned / enrollment stuck" is diagnosed by confirming the Mac's **serial** is assigned to your MDM server in ABM/ASM and that the **enrollment program token** in Intune is current. That assignment lives in Apple's portal, not Intune. **Verify:** Sign in to `https://business.apple.com` → **Devices**; you can search a serial and see its assigned **MDM Server**. In Intune → **Devices** → **Enrollment** → **Enrollment program tokens**, the token shows a future expiry and a set **default enrollment profile**. **Implement:** Obtain an ABM/ASM administrator or Device Enrollment Manager role; if you cannot get it, identify the person who can and include them in the runbook. Renew the token before expiry (2.7).

2.5 On-device diagnostic access (the failing Mac)

Description: Physical or remote (admin) access to the affected Mac, with an admin account and Terminal, plus the ability to read the on-Mac state. **Why:** SecureToken/FileVault status and SSO-extension registration are read **on the Mac** with built-in tools — Graph can confirm escrow and profile delivery, but the local token state is authoritative on the device. **Verify:** On the Mac you can run, as admin: `sysadminctl -secureTokenStatus theusername`, `fdesetup status`, `profiles status -type enrollment`, and `app-sso platform -s` (Platform SSO) — each returns output, not "command not found"/permission errors. **Implement:** Ensure an admin account exists on the Mac (or use a remote-support tool that grants an admin shell); document the built-in commands in your runbook so any technician can run them.

2.6 Device logs and diagnostics

Description: The ability to collect macOS enrollment/MDM logs and, where needed, the **Company Portal** / SSO extension diagnostics. **Why:** When a profile shows `Error` in Intune, the reason is often in the Mac's MDM log; when SSO loops, the SSO extension log names the failing step. **Verify:** On the Mac, **Console.app** shows `mdmclient`/`profiles` activity, or `log show --predicate 'subsystem == "com.apple.ManagedClient"' --last 1h` returns entries; Company Portal → **Help** → **Save diagnostic report** produces a bundle. **Implement:** Enable technician access to Console/`log`; teach the Company Portal diagnostic-report path; keep a folder for collected logs per ticket.

2.7 Current enrollment program token and profiles

Description: A non-expired Apple enrollment program token in Intune, a set default enrollment profile, and the required macOS configuration profiles (SSO extension, FileVault, compliance-supporting) published and assigned. **Why:** An expired token stops all new ADE enrollments tenant-wide; a missing default profile means devices enroll without your settings; undelivered profiles break the SSO and FileVault layers. **Verify:** Intune → **Devices** → **Enrollment** → **Enrollment program tokens** → token expiry is in the future and a **default profile** is assigned; **Devices** → **Configuration** lists the SSO-extension, FileVault, and compliance-supporting profiles with a macOS platform and a group assignment. **Implement:** Renew the token from ABM/ASM before expiry; set a default enrollment profile; publish and assign the SSO-extension (Platform/Extensible SSO), FileVault, and compliance-supporting profiles to the pilot group.

2.8 Supported macOS baseline

Description: A supported macOS version on the affected devices (Apple's current and two prior major releases are the safe target; Platform SSO requires macOS 13+ / 14+ features). **Why:** SecureToken/Bootstrap-Token behavior and Platform SSO differ by macOS major version; diagnosing on an unsupported build wastes time. **Verify:** On the Mac, `sw_vers`; in Intune, device → **Hardware** → **Operating system**. **Implement:** Update macOS to a supported release before deep troubleshooting; confirm the SSO method you use is supported on that version.

2.9 A reproducible failure to pilot on

Description: At least one real, current failing Mac (or a device you can re-run through Setup Assistant) for each symptom class you intend to validate. **Why:** The playbook is validated by diagnosing actual failures — not by reading it. **Verify:** Intune → **Devices** → **macOS** shows a device in the failing state (not enrolled, non-compliant, profile in Error, etc.), or you have a wipeable test Mac. **Implement:** Use a currently failing device; if none exists, erase-and-reenroll a spare Mac to reproduce an enrollment/SSO/FileVault symptom safely.

Checklist

- Intune admin center loads; macOS devices, Enrollment, Configuration, Compliance visible
- `Microsoft.Graph.Authentication` installed; read scopes consented; `Get-MgContext` works
- Intune Administrator role assigned; Entra Sign-in logs readable

- ABM/ASM access (or a named contact); enrollment token current with a default profile
- Admin/Terminal access to the failing Mac; built-in diagnostic commands run
- macOS MDM/SSO logs and Company Portal diagnostics collectable
- SSO-extension, FileVault, and compliance-supporting profiles published and assigned
- Supported macOS baseline confirmed for the SSO method in use
- A real failure identified (per symptom class) to pilot the playbook on

3. Diagnostic Playbook — macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit

This is the runbook. It takes a stuck Mac, classifies the symptom, and drives it to the failing layer and a fix. Console work is in the **Microsoft Intune admin center** (`https://intune.microsoft.com`); on-device checks run in **Terminal** on the affected Mac (admin); the scripts run from a PowerShell session with the **Microsoft.Graph.Authentication** module signed in with the read scopes (§2.2).

> ***Method:*** *Classify the symptom → confirm the layer **above** it is healthy (top-down) → read the layer's signal → apply the branch fix → re-check → record. Name every artifact you create with* `<Purpose> – <Scope> – <Stage>` *and reuse the names (e.g.* `macOS SSO Fix – Pilot – Triage`*).*

Step 0 — Classify the symptom and gather context

1. Ask the one question that picks the branch: **where does it stop?**

- No Remote Management / enrollment never finishes → **Branch A (ADE/enrollment)**.
- Enrolled, but the SSO prompt never works → **Branch B (SSO)**.
- FileVault won't enable or a user can't unlock the disk → **Branch C (SecureToken/FileVault)**.
- Device shows Compliant but a resource is blocked → **Branch D (compliant but denied)**.
- Enrolled, but the user can't log in / password is wrong → **Branch E (locked out)**.

1. Get the device context first — run `MACDX-Discovery.ps1` to list the macOS device, its enrollment state, ownership, compliance, last check-in, and OS build. This confirms which layer the device has reached and gives you the device id the other scripts use.

Branch A — ADE not assigned / enrollment stuck

Tell: Setup Assistant shows no **Remote Management** screen, or enrollment hangs/errors. **Diagnose top-down (Apple assignment → Intune token → profile):**

1. **Serial assigned in Apple.** In **Apple Business Manager / School Manager** → **Devices**, search the Mac's serial. Confirm it is assigned to **your** MDM server (the Intune server). If it isn't there, the device was never bought through / added to ABM/ASM — no ADE is possible.

1. **Token healthy in Intune.** Intune → **Devices** → **Enrollment** → **Enrollment program tokens**. Confirm

the token's **expiry is in the future** and a **default enrollment profile** is set. Run `MACDX-Validation.ps1 -Check Enrollment` to read the token/profile summary via Graph.

1. **Profile assigned to the device.** In the token → **Profiles** and **Devices**, confirm the Mac's

serial has an enrollment profile assigned (not just the default). Re-sync ABM/ASM (**Sync**) if the serial is new.

1. **On the Mac.** After erase, at Setup Assistant, or via `sudo profiles renew -type enrollment`

(macOS reaches out again). `profiles status -type enrollment` should report it is enrolled / has a configuration waiting.

Fix:

- Serial not assigned → have the reseller/Apple assign it, or add it to ABM/ASM and **Sync** to Intune.
- Token expired → **renew** the token (download a new one from ABM/ASM, upload to Intune).
- No default profile → set a **default enrollment profile** on the token and assign it to the serial.
- Stuck mid-enrollment → erase and re-run Setup Assistant, or `sudo profiles renew -type enrollment`.

Confirm: The Mac reaches **Remote Management** and appears in Intune → **Devices** → **macOS** as enrolled. Re-run `MACDX-Validation.ps1`.

Branch B — SSO won't sign in (Platform SSO / Extensible SSO)

Tell: The Platform/Extensible SSO prompt never appears, loops, or rejects the Entra credential. **Diagnose top-down (profile delivered → app installed → extension registered → method correct):**

1. **Extension profile delivered.** Intune → device → **Device configuration**; the **SSO app

extension **(Platform SSO or the Microsoft Enterprise SSO / Extensible SSO) profile shows** Succeeded**, not Error/Conflict. Run `MACDX-Validation.ps1 -Check Profiles` to list per-device profile states via Graph.

1. **Required app installed.** The extension needs its companion app present and run once:

Company Portal (Platform SSO / Company Portal-based registration) or Microsoft **Authenticator** (Extensible SSO plug-in). Confirm the app is assigned (Intune → **Apps** → **macOS**) and installed on the Mac.

1. **Extension registered on the Mac.** In Terminal: `app-sso platform -s` (Platform SSO) shows the

registration/state; `app-sso -l` lists SSO extensions. For Extensible SSO, **System Settings** → **Profiles** shows the SSO extension profile installed.

1. **Method correct.** Confirm the configured Platform SSO **authentication method** (Password,

Secure Enclave key, or Smart Card) matches what the user has and what the profile specifies; a Secure-Enclave-key config with no registered key loops.

Fix:

• Profile in Error/Conflict → resolve the conflicting profile / re-assign; confirm the Mac is in the

target group (`MACDX-Set-ScopeTargeting.ps1`, `-WhatIf` by default).

• App missing → assign/install Company Portal or Authenticator; open it once to register.

• Not registered → have the user complete registration in Company Portal; `sudo profiles renew` to

re-pull the extension; re-run after check-in.

• Wrong method → align the profile's Platform SSO method with the user's capability; re-deploy.

Confirm: The user signs in through SSO and `app-sso platform -s` shows a registered/logged-in state. Re-run `MACDX-Validation.ps1 -Check Profiles`.

Branch C — SecureToken / FileVault error

Tell: FileVault won't enable, a user can't unlock the disk at boot, or the recovery key isn't escrowed.
Diagnose top-down (SecureToken holder → Bootstrap Token escrow → FileVault profile → key escrow):

1. **SecureToken holder.** On the Mac: `sysadminctl -secureTokenStatus theusername` for the affected user

and the first admin. A user with **ENABLED** SecureToken can be a FileVault user; a user with **DISABLED** cannot enable/unlock FileVault.

1. **Bootstrap Token escrowed.** `profiles status -type bootstraptoken` shows whether the Bootstrap

Token is escrowed to the MDM. Modern Apple-silicon FileVault relies on the Bootstrap Token; if it was never escrowed, MDM-driven enablement and additional-user grants fail.

1. **FileVault profile state.** Intune → device → **Device configuration**; the FileVault

(Endpoint security → Disk encryption) policy shows **Succeeded**. Run `MACDX-Validation.ps1 -Check FileVault`.

1. **Recovery key escrowed.** Intune → device → **Recovery keys** shows a stored **personal recovery

key**. `MACDX-Report.ps1` includes an escrow-coverage count across the fleet. `fdesetup status` on the Mac shows whether FileVault is On.

Fix:

• User lacks SecureToken → grant it from a SecureToken-holder admin:

`sysadminctl -secureTokenOn theusername -password userpassword -adminUser adminname -adminPassword adminpassword` (run on the Mac; substitute the real account names and passwords).

• Bootstrap Token not escrowed → `sudo profiles install -type bootstraptoken` (or re-enroll) to

escrow it; confirm with `profiles status -type bootstraptoken`.

• FileVault profile in Error → fix the assignment/settings and re-deploy; ensure the target account

holds SecureToken first.

• Key not escrowed → rotate/re-escrow the personal recovery key via the FileVault policy; confirm it

appears under **Recovery keys**.

Confirm: `fdesetup status` shows FileVault On for the intended user, the recovery key is visible under **Recovery keys** in Intune, and the user unlocks the disk at boot. Re-run `MACDX-Validation.ps1 -Check FileVault`.

Branch D — Compliant but denied

Tell: Intune shows the Mac **Compliant**, yet the user is blocked from a resource (M365, etc.). **Diagnose top-down (compliance signal → Entra sign-in log → Conditional Access policy):**

1. **Compliance is real and current.** Intune → device → **Compliance**; the policy shows

Compliant with a recent evaluation. Run `MACDX-Validation.ps1 -Check Compliance` — it names any non-compliant setting. A device that flipped compliant seconds ago may not have propagated yet.

1. **Read the actual block.** Entra → **Sign-in logs** → filter to the user and the time of the block

→ open the failed sign-in → **Conditional Access** tab. It names the **policy** that applied and the **grant control** that failed (e.g. *Require device to be marked as compliant*, *Require approved app*, *Require hybrid Entra join*).

1. **Device identity match.** Confirm the device Entra ID in the sign-in log matches the compliant

Intune device — a second/duplicate device record, or a browser not using the SSO extension, can present as a *different* (non-compliant/unknown) device to CA.

Fix:

• Signal lag → wait for propagation, or force a device **Sync** (`MACDX-Remediation.ps1 -Action Sync`,

`-WhatIf` by default) and re-test after check-in.

• CA grant the Mac can't meet → adjust the policy scope/exclusion or give the Mac the required state

(e.g. ensure the SSO extension is delivering device compliance to the browser — Branch B).

• Duplicate/stale device object → reconcile the duplicate Entra device record so CA evaluates the

compliant one.

Confirm: The same sign-in now shows **Success** with the CA policy **Satisfied** in Entra Sign-in logs.

Branch E — User locked out after enrollment

Tell: Enrollment succeeded, but the user can't log in, is prompted for an unknown password, or the local account is out of sync with Entra. **Diagnose top-down (account model → Platform SSO binding → FileVault account → password sync):**

1. **Which account model.** Determine whether the Mac uses **Platform SSO** account binding (local

account linked to Entra), a **mobile account**, or a plain local admin. `app-sso platform -s` and `dscl . -read /Users/theusername` show the account state.

1. **Platform SSO binding health.** If Platform SSO is in use, confirm the extension is registered and

the user completed sign-in (Branch B) — a half-bound account can reject the Entra password locally.

1. **FileVault account match.** If FileVault is On, the login/unlock account must be a

SecureToken-holding, FileVault-enabled user (Branch C). A user who isn't FileVault-enabled can't unlock at boot even with the right password.

1. **Password sync.** With Platform SSO password mode, a recent Entra password change must sync; the

user may need to sign in with the **previous** local password once, then the extension syncs.

Fix:

• Half-bound Platform SSO → re-register via Company Portal; `sudo profiles renew` and re-run sign-in.

• FileVault account mismatch → grant SecureToken and add the user to FileVault (Branch C).

• Password out of sync → follow the Platform SSO password-sync path (sign in local, let it sync); as a

last resort an admin resets the local password from a SecureToken holder.

• No admin available on the box → use the escrowed **personal recovery key** (Intune → **Recovery

keys**) to unlock, then repair the account.

Confirm: The user logs in and unlocks the disk with their Entra credential (or synced password), and `app-sso platform -s` shows a healthy registered state.

Step 9 — Record the resolution

Capture the symptom class, the failing layer, the signal you read, the root cause, and the fix in your runbook log (name it `macOS Failure Log – <Site> – Runbook`). This is what makes the next ticket minutes instead of hours.

Step 10 — Validate and expand

1. Run the full loop on **at least two** distinct real failures (ideally different branches); confirm

each was isolated to the correct layer and resolved. Complete the **Testing Guide** (§5).

1. Save the scripts, the recorded resolutions, and the on-Mac command cheat-sheet where the team

works. Make this the standard triage path for macOS tickets; re-validate as macOS versions and the SSO method change.

Order of operations

`Classify symptom → confirm layer above (top-down) → read the layer's signal → apply branch fix → re-check the signal → record → validate on real failures`

→ `expand`

5. Testing Guide — macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit

Run these after you have read access set up and have piloted the playbook on a real failure (Diagnostic Playbook §3). These tests prove the toolkit actually isolates and resolves the failing layer — not that the doc reads well. Each test maps to a symptom branch.

Test 1 — Device context is visible (Discovery)

Do: Run `MACDX-Discovery.ps1` (optionally `-DeviceName device-name-or-serial`). **Expected:** The macOS device is listed with enrollment state, ownership, compliance, OS build, and last check-in — read-only, with no attempt to change the tenant. PASS if it connects with the read scopes and lists devices; FAIL if sign-in or scopes are missing.

Test 2 — Enrollment / ADE state (Branch A)

Do: Run `MACDX-Validation.ps1 -DeviceName device-name-or-serial -Check Enrollment`; cross-check Intune → **Devices** → **Enrollment** → **Enrollment program tokens**. **Expected:** The script reports the enrollment token's expiry and whether a default profile is set, consistent with the portal; a stuck device is shown as not-enrolled with the reason class visible.

Test 3 — SSO extension state (Branch B)

Do: Run `MACDX-Validation.ps1 -DeviceName device-name-or-serial -Check Profiles`; on the Mac run `app-sso platform -s`. **Expected:** The SSO app-extension profile state (`Succeeded`/`Error`/`Conflict`) matches the portal, and the on-Mac registration state agrees. A failing SSO device shows the profile not `Succeeded` or the extension not registered.

Test 4 — SecureToken / FileVault state (Branch C)

Do: Run `MACDX-Validation.ps1 -DeviceName device-name-or-serial -Check FileVault`; on the Mac run `sysadminctl -secureTokenStatus theusername` and `fdesetup status`. **Expected:** The FileVault profile state and recovery-key escrow reported by the script match Intune → device → **Recovery keys**; the on-Mac SecureToken/FileVault status agrees. A failing device shows no escrowed key or a SecureToken-DISABLED account.

Test 5 — Compliance / Conditional Access (Branch D)

Do: Run `MACDX-Validation.ps1 -DeviceName device-name-or-serial -Check Compliance`; open Entra → **Sign-in logs** → **Conditional Access** for the blocked sign-in. **Expected:** The script names any non-compliant setting; the sign-in log names the CA policy and the grant control that denied. A "compliant but denied" case shows Compliant in Intune yet a failed CA grant in Entra.

Test 6 — Fleet report and escrow coverage

Do: Run `MACDX-Report.ps1`. **Expected:** A clean summary — counts by enrollment/compliance state, stale (not-checked-in) macOS devices, OS-version spread, and FileVault recovery-key escrow coverage — reconciling with the Intune portal for your scope.

Test 7 — Safe remediation preview (default -WhatIf)

Do: Run `MACDX-Remediation.ps1 -DeviceName device-name-or-serial -Action Sync` **without** `-WhatIf:$false`. **Expected:** It prints the intended safe action (device Sync) and makes **no** change (PREVIEW MODE). Re-running with `-WhatIf:$false` on a pilot device issues the Sync and reports success.

Test 8 — Scope targeting preview (default -WhatIf)

Do: Run `MACDX-Set-ScopeTargeting.ps1 -GroupName pilot-group-name -DeviceName device-name-or-serial` **without** `-WhatIf:$false`. **Expected:** It prints the intended group and membership change and makes **no** change. With `-WhatIf:$false` on a pilot group it creates the group if needed and adds the device.

Test 9 — Re-check after fix (any branch)

Do: After applying a branch fix, re-run the matching `MACDX-Validation.ps1 -Check ...` and the on-Mac command for that layer. **Expected:** The signal that was failing now reads healthy (enrolled / SSO registered / FileVault On + key escrowed / CA satisfied), and the symptom is gone.

Pass/fail gate

Gate	Requirement
Read pass	Tests 1–6 pass read-only: device context, per-branch state, and the fleet report reconcile with the Intune portal and Entra sign-in logs
Safety pass	Tests 7–8 pass: Remediation and Set-ScopeTargeting default to `-WhatIf` and make no change until `-WhatIf:$false`
Resolution pass	Test 9 passes on **two** distinct real failures (Playbook §3, Step 10): each was isolated to the correct layer and the re-check reads healthy

6. Troubleshooting Guide — macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit

This is the formal troubleshooting section for the **toolkit and its tooling** — what to do when the diagnostic process itself misbehaves (Graph sign-in fails, a device isn't found, an on-Mac command is denied, escrow reads empty). For diagnosing the *Mac failure* itself, follow the Diagnostic Playbook (§3) and its five branches.

A. Microsoft Graph sign-in / scope errors

Symptom	Cause	Remediation
`Connect-MgGraph` fails or module missing	`Microsoft.Graph.Authentication` not installed	`Install-Module Microsoft.Graph.Authentication -Scope CurrentUser`, then retry
`Authorization_RequestDenied` on a read	The consented scopes don't include the one the check needs	Reconnect with the read scopes in §2.2; an admin may need to consent
Remediation/Targeting write is refused	Missing write scope	Reconnect adding `DeviceManagementManagedDevices.PrivilegedOperations.All` (Sync) or `Group.ReadWrite.All`/`GroupMember.ReadWrite.All` (targeting)
Wrong tenant	Cached context points elsewhere	`Disconnect-MgGraph`, then reconnect with `-TenantId your-tenant-id`

B. Device not found by the scripts

• The scripts match macOS managed devices by **device name or serial** (substring). If nothing matches, confirm the device is enrolled (Intune → **Devices** → **macOS**) and try the **serial** — a freshly enrolled Mac may not have its friendly name yet.

• If `MACDX-Discovery.ps1` lists the device but `-DeviceName` in another script misses it, use a more specific substring (the full serial) so it isn't ambiguous.

C. On-Mac commands are denied or "command not found"

Symptom	Cause	Remediation
`sysadminctl`/`fdesetup` "not permitted"	Not running as an admin, or SecureToken-holder needed	Run in Terminal as a local admin; SecureToken grants need a SecureToken-holding admin's credentials
`app-sso platform -s` returns nothing	Platform SSO not configured / older macOS	Confirm the SSO method is Platform SSO and macOS supports it (§2.8); for Extensible SSO check **System Settings** → **Profiles** instead
`profiles status -type bootstraptoken` "not escrowed"	Bootstrap Token was never escrowed	`sudo profiles install -type bootstraptoken` or re-enroll (Playbook Branch C)

D. Profile shows Error/Conflict in Intune but you can't tell why

• Two profiles setting the **same** payload differently produce a **Conflict** — find the second

profile targeting the device and reconcile them; the Mac's group membership tells you which profiles apply (`MACDX-Set-ScopeTargeting.ps1` preview shows current membership).

- For an **Error**, read the Mac's MDM log: `log show --predicate 'subsystem == "com.apple.ManagedClient"' --last 1h`

or Console.app for the `mdmclient`/`profiles` entry that names the failing payload.

E. Recovery keys / escrow reads empty

- Intune → device → **Recovery keys** blank while `fdesetup status` shows FileVault On → the

personal recovery key wasn't escrowed. Rotate/re-escrow via the FileVault (Disk encryption) policy; the key appears after the next check-in.

- `MACDX-Report.ps1` escrow coverage shows 0 → confirm the FileVault policy is assigned and the

reading account has `DeviceManagementConfiguration.Read.All` / `DeviceManagementManagedDevices.Read.All`.

F. "Compliant but denied" doesn't resolve

- If Intune shows Compliant but Entra keeps denying, the block is a **Conditional Access** decision,

not an Intune one — read the CA tab in the **Sign-in log**, not the compliance blade. The named policy + grant control is the actual cause (Playbook Branch D).

- A browser that doesn't use the SSO extension can present as a different/unknown device to CA even

though the OS is compliant — confirm the SSO extension is delivering device state (Branch B).

G. Remediation / targeting "made no change"

- Both `MACDX-Remediation.ps1` and `MACDX-Set-ScopeTargeting.ps1` default to `-WhatIf` — that is

by design. They only act when you re-run with `-WhatIf:$false`. The `[PREVIEW MODE]` banner tells you no change was made.

- A Sync issued but nothing changed on the Mac → allow a few minutes for the device to check in, then

re-run `MACDX-Validation.ps1` for that layer.

H. Diagnostics quick reference

```
# Confirm the Graph context and scopes the scripts will use
Get-MgContext | Select-Object Account, TenantId, Scopes

# List macOS managed devices (read-only), newest check-in first
Invoke-MgGraphRequest -Method GET `
  -Uri "https://graph.microsoft.com/v1.0/deviceManagement/managedDevices?`$filter=operatingSystem eq 'macO
  Select-Object -ExpandProperty value |
  Select-Object deviceName, serialNumber, complianceState, osVersion, lastSyncDateTime |
  Sort-Object lastSyncDateTime -Descending | Select-Object -First 20

# On the affected Mac (Terminal, as admin) — the four layer signals
sudo profiles status -type enrollment          # ADE / enrollment (Branch A)
app-sso platform -s                            # Platform SSO registration (Branch B)
```

```
sysadminctl -secureTokenStatus theusername     # SecureToken holder (Branch C)
sudo profiles status -type bootstraptoken      # Bootstrap Token escrow (Branch C)
fdesetup status                                # FileVault On/Off (Branch C)
```

I. When to escalate

If the layer signal is read correctly, the branch fix is applied, and the symptom persists, collect: the Company Portal **diagnostic report**, the Mac's MDM log (`log collect --last 2h` / Console.app), the Intune device page, and the Entra **Sign-in log** for the failed CA case. Open a Microsoft support case with the symptom class, the failing layer, and the exact signal you captured — and, for an Apple-side ADE assignment issue, engage Apple/ABM support with the serial and the assigned MDM server.

7. Best Practices — macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit

Evidence-based escalation: every recommendation below shortens the path from symptom to the failing layer, and keeps you from acting on a guess.

- **Diagnose top-down, always.** A lower-layer symptom (SSO, FileVault, CA) is usually caused by a

missing or errored artifact **above** it (an unassigned serial, an undelivered profile). Confirm the higher layer is healthy before you blame the lower one — it is the single biggest time-saver.

- **Classify the symptom first.** The one question "**where does it stop?**" picks the branch. Naming

the class (ADE / SSO / FileVault / compliant-but-denied / locked-out) before touching anything keeps the investigation on rails.

- **Read the signal; don't infer it.** Each layer has an authoritative signal — the enrollment token,

the per-device profile state, `sysadminctl`/`fdesetup` on the Mac, the Entra sign-in log's CA tab. Read the actual signal rather than assuming from the symptom.

- **Trust the Mac for token state.** SecureToken and Bootstrap-Token status are authoritative **on the

device** (`sysadminctl`, `profiles status -type bootstraptoken`). Graph confirms escrow and profile delivery; the local commands confirm the token itself.

- **Escrow the recovery key before you need it.** Verify FileVault **personal recovery key** escrow in

Intune (and `MACDX-Report.ps1` coverage) proactively — the key is what unlocks a locked-out Mac in Branch E. Missing escrow turns a five-minute fix into a wipe.

- **Keep the enrollment token current.** An expired Apple enrollment program token stops **all** new

ADE enrollments tenant-wide. Track the expiry and renew ahead of time; set and verify a default enrollment profile.

- **Preview before you change.** Run `MACDX-Remediation.ps1` and `MACDX-Set-ScopeTargeting.ps1` in

their default `-WhatIf` mode, read the intended action, then re-run with `-WhatIf:$false`. The safe actions here (Sync, add-to-group) are deliberately the least-disruptive; never reach for a wipe as a first move.

• **Separate "compliant" from "allowed".** Intune decides compliance; **Conditional Access** decides

access. When a compliant Mac is denied, read the CA tab of the Entra sign-in log — that names the policy and grant control actually blocking, so you fix the right thing.

• **Confirm the SSO method matches the device.** Platform SSO password vs. Secure-Enclave-key vs.

smart-card each has different device prerequisites; a mismatch loops sign-in. Verify macOS supports the method you configured before deploying it fleet-wide.

• **Record every resolution.** A short log of symptom → failing layer → signal → cause → fix compounds:

the second occurrence should be minutes, not hours, and it is what lets you hand the runbook to the rest of the team.

• **Pilot on real failures.** Validate the playbook on at least two distinct failures — ideally

different branches — before making it the team standard. Assign once, repeat everywhere.

• **Escalate with evidence, not a story.** When a fix doesn't hold, escalate with the captured

signal — Company Portal diagnostics, the MDM log, the Intune device page, the Entra CA sign-in — and the exact layer/class. Evidence-based tickets get resolved; "it won't enroll" tickets bounce.

8. Deliverables Summary — macOS Enrollment, SSO & FileVault Troubleshooting & Diagnostics Toolkit

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	The macOS + Intune signal path (ADE/enrollment → profiles → SSO → SecureToken/FileVault → compliance/CA), the layered model, the five failure classes, and the diagnose-then-resolve method
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing — Intune + Graph read access, ABM/ASM vantage point, on-Mac diagnostic access, device logs
Global Requirements (Infrastructure)	Category-wide prerequisites, published once
3. Diagnostic Playbook	The symptom tree: classify → confirm the layer above → read the signal → branch fix, across all five branches (ADE/enrollment, SSO, SecureToken/FileVault, compliant-but-denied, locked-out)
5. Testing Guide	Validation procedures with pass/fail gates, one per branch
6. Troubleshooting Guide	Troubleshooting the toolkit itself (Graph sign-in, device lookup, on-Mac commands, escrow reads)

Document	Purpose
7. Best Practices	Evidence-based escalation and field-proven recommendations

PowerShell scripts (Microsoft Graph SDK)

Script	What it does
`MACDX-Discovery.ps1`	Lists macOS managed devices with enrollment state, ownership, compliance, OS build, and last check-in (read-only)
`MACDX-Validation.ps1`	Per-device, per-layer state check (`-Check Enrollment/Profiles/FileVault/Compliance`): enrollment/token, SSO-extension profile state, FileVault + recovery-key escrow, named non-compliant settings (read-only)
`MACDX-Report.ps1`	Fleet summary — enrollment/compliance mix, stale devices, OS-version spread, FileVault recovery-key escrow coverage (read-only)
`MACDX-Remediation.ps1`	Safe remediation helper (`-WhatIf` by default): issue a device **Sync** to re-pull policy and re-evaluate; never wipes/resets/retires
`MACDX-Set-ScopeTargeting.ps1`	Pilot/target-group builder (`-WhatIf` by default): ensure an Entra security group exists and add the device so a missing profile/app flows to it

What the buyer receives

- This playbook as a **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP (Graph SDK; read-only Discovery/Validation/Report;

Remediation and Set-ScopeTargeting default to `-WhatIf`).

- A repeatable process following the standard model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

The failing layer of a stuck Mac isolated in minutes — enrollment, SSO, SecureToken/FileVault, or compliance — by classifying the symptom, confirming top-down, and reading the one authoritative signal per layer. A repeatable diagnostic capability the team keeps and reuses, validated on real failures before it scales out.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

MACDX-Discovery.ps1

Purpose: Discovers the macOS + Intune fleet so a technician can triage at scale instead of clicking each device. Pulls managed macOS device inventory, ownership, enrollment/compliance state, last check-in, and OS build from Microsoft Graph. Read-only: it never changes the tenant. This is Step 0 of the Diagnostic Playbook — see which layer a device has reached and get the device id the other scripts use. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\MACDX-Discovery.ps1 powershell -ExecutionPolicy Bypass -File .\MACDX-Discovery.ps1 -DeviceName device-name-or-serial -StaleDays 14 -CsvPath .\macos-devices.csv

How to run: powershell -ExecutionPolicy Bypass -File .\MACDX-Discovery.ps1 powershell -ExecutionPolicy Bypass -File .\MACDX-Discovery.ps1 -DeviceName device-name-or-serial -StaleDays 14 -CsvPath .\macos-devices.csv

MACDX-Remediation.ps1

Purpose: Applies the safest, least-disruptive fix the Diagnostic Playbook calls for: trigger an Intune device Sync so the Mac re-pulls its enrollment profile, configuration profiles (incl. the SSO extension and FileVault policy), and re-reports compliance. It reads device state from Microsoft Graph, then issues only the low-risk Sync action. It defaults to -WhatIf and makes no change until you re-run with -WhatIf:$false. It never wipes, erases, resets, retires, or deletes anything. HOW TO RUN Preview (no change, default): powershell -ExecutionPolicy Bypass -File .\MACDX-Remediation.ps1 -DeviceName device-name-or-serial Apply after reviewing: powershell -ExecutionPolicy Bypass -File .\MACDX-Remediation.ps1 -DeviceName device-name-or-serial -WhatIf:$false

How to run: Preview (no change, default): powershell -ExecutionPolicy Bypass -File .\MACDX-Remediation.ps1 -DeviceName device-name-or-serial Apply after reviewing: powershell -ExecutionPolicy Bypass -File .\MACDX-Remediation.ps1 -DeviceName device-name-or-serial -WhatIf:$false

MACDX-Report.ps1

Purpose: Produces a macOS fleet summary a lead can use to spot trends and coverage gaps: counts by enrollment and compliance state, stale (not-checked-in) Macs, OS-version spread, and the FileVault personal-recovery-key escrow coverage across the fleet. Drawn from Microsoft Graph. Read-only: it never changes the tenant. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\MACDX-Report.ps1 powershell -ExecutionPolicy Bypass -File .\MACDX-Report.ps1 -StaleDays 14 -CsvPath .\macos-fleet.csv

How to run: powershell -ExecutionPolicy Bypass -File .\MACDX-Report.ps1 powershell -ExecutionPolicy Bypass -File .\MACDX-Report.ps1 -StaleDays 14 -CsvPath .\macos-fleet.csv

MACDX-Set-ScopeTargeting.ps1

Purpose: Fixes the group-based targeting the Diagnostic Playbook relies on: ensures a pilot/target Entra security group exists and adds the named macOS device (its Entra device object) to it, so a device starts receiving the SSO-extension, FileVault, or compliance-supporting profile it was missing. It reads current

membership from Microsoft Graph and only adds the member you name. It defaults to -WhatIf and makes no change until you re-run with -WhatIf:$false. It never removes members or deletes groups. HOW TO RUN Preview (no change, default): powershell -ExecutionPolicy Bypass -File .\MACDX-Set-ScopeTargeting.ps1 -GroupName pilot-group-name -DeviceName device-name-or-serial Apply after reviewing: powershell -ExecutionPolicy Bypass -File .\MACDX-Set-ScopeTargeting.ps1 -GroupName pilot-group-name -DeviceName device-name-or-serial -WhatIf:$false

How to run: Preview (no change, default): powershell -ExecutionPolicy Bypass -File .\MACDX-Set-ScopeTargeting.ps1 -GroupName pilot-group-name -DeviceName device-name-or-serial Apply after reviewing: powershell -ExecutionPolicy Bypass -File .\MACDX-Set-ScopeTargeting.ps1 -GroupName pilot-group-name -DeviceName device-name-or-serial -WhatIf:$false

MACDX-Validation.ps1

Purpose: Validates the state of one macOS device at the layer a Diagnostic Playbook branch cares about, so a technician can confirm what a check expects. -Check selects the layer: Enrollment (ADE token/profile context), Profiles (SSO app-extension and other profile states), FileVault (disk-encryption profile state + personal recovery-key escrow), or Compliance (named non-compliant settings). Read-only: it never changes the tenant. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\MACDX-Validation.ps1 -DeviceName device-name-or-serial -Check Profiles powershell -ExecutionPolicy Bypass -File .\MACDX-Validation.ps1 -DeviceName device-name-or-serial -Check FileVault

How to run: powershell -ExecutionPolicy Bypass -File .\MACDX-Validation.ps1 -DeviceName device-name-or-serial -Check Profiles powershell -ExecutionPolicy Bypass -File .\MACDX-Validation.ps1 -DeviceName device-name-or-serial -Check FileVault